

Max Rodriguez

Software Engineer | AI Systems & Applied Research

maxrod@stanford.edu | Portfolio | GitHub | LinkedIn

EDUCATION

M.S. Computer Science, Artificial Intelligence | Expected 2027

Stanford University | GPA 3.865 / 4.00

B.S. Symbolic Systems, Artificial Intelligence, with Honors | 2026

Stanford University | GPA 3.974 / 4.00 | Minors: Music and German Studies

EXPERIENCE

Bonterra | Jun - Sep 2026

AI Software Engineer Intern | Stanford CS + Social Good Fellow

- Built TypeScript/React AI tools for nonprofit software workflows, including an agentic pre-call briefing system over Salesforce and Gong with role-specific briefs and attributable sources.
- Implemented a live call-coaching agent; validated behavior with unit, integration, and security tests and translated stakeholder discovery into product requirements.

WAW Life, Inc. | Jun 2025 - Present

Software Engineering & Product Lead

- Designed and built wawbeverage.com with React, TypeScript, Vite, a custom GLSL hero, and Shopify checkout.
- Built and deployed a Next.js/Supabase VIP-ordering PWA with magic-link authentication, customer/admin roles, fulfillment tooling, and Make.com webhook order routing.

E3 Group | Jun - Jul 2026

Software Engineer, Contract

- Built multi-stage driver-license verification workflows combining OCR, barcode decoding, and LLM extraction; developed image augmentation and stage-level benchmarks to diagnose extraction failures.

Stanford Social Media Lab | Jun 2025 - Present

Research Assistant & AI Systems Engineer | CURIS Fellow

- Benchmarked four feed-ranking architectures across 169,787 target interactions and 8,214 users; best fully interacted two-tower model reached 0.8339 holdout AUC.
- Trained a PyTorch image-text classifier 1.94x faster than a scikit-learn baseline; engineered a Torch/PIL anonymization pipeline over 100,000+ posts.

Technische Universität Berlin | Jun - Sep 2024

Functional Sound Research / App Development Intern

- Built Somunicate, a Python/Streamlit recommender matching UX sounds to perceptual inputs with Mahalanobis distance.
- Built a watermarking app for copyrighted sounds; collaborated with the Audio Communication Group on UX-sound research.

RESEARCH & PROJECTS

FiMo-Klavier | Published Stanford Honors Thesis

Sole author of a video-only piano technique feedback system spanning 137 hours and 1.02M motion windows: MediaPipe features, BiLSTM-Transformer anomaly detection, 114 Gaussian HMMs, and gated LLM feedback. DOI: [10.25740/vk447bc1950](https://doi.org/10.25740/vk447bc1950).

Hybrid Advantage Shaping | LLM-Agent Reinforcement Learning

Co-developed goal-conditioned per-turn credit assignment. Team results improved WebShop success from 80.5% to 91.3% and ALFWorld from 70% to 80% over flatGRPO (Qwen2.5-1.5B, rank-32 LoRA).

Evensong | Independent Game Development

Designing and developing a narrative pixel-art adventure in Godot/C#: persistent world state, Ink dialogue, environment presentation, and automated story-validation tools. In development.

SKILLS & RECOGNITION

Engineering: Python, TypeScript/JavaScript, C/C++, C#, SQL, React, Next.js, Node.js, FastAPI, Supabase, Docker, Git

AI/ML: PyTorch, Transformers, Hugging Face, LoRA, PPO/GRPO/DAPO, retrieval, computer vision, OCR, LLM agents

Practice & honors: Testing, benchmarking, failure analysis | Cardinal Quarter Fellow, CS147 Best Demo, HSF Scholar